COMPARATIVE MULTI-SATELLITE STATISTICAL ANALYSIS OF GLACIER RETREAT & MORPHOLOGY

A Harmonized High-Resolution Remote-Sensing Evaluation of Gangotri Glacier in 2025

Academic Context: Minor Project (B.Tech Information Technology, 7th Semester)
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Date of Evaluation: Ablation Season 2025 | **Target Architecture:** 8-Channel Attention U-Net

ABSTRACT — *Systematic monitoring of Himalayan cryospheric dynamics is essential for water security and glacial lake outburst flood (GLOF) hazard mitigation. While deep learning architectures like the 8-Channel Attention U-Net have revolutionized automated glacier boundary delineation, their operational reliability across divergent orbital sensors remains a critical open question. This research conducts a rigorous comparative statistical analysis between Copernicus Sentinel-2 (MSI) and Landsat 8/9 (OLI) observations over the Gangotri Glacier basin (Garhwal Himalaya, Uttarakhand) during the 2025 ablation season. Using NASA Harmonized Landsat Sentinel-2 (HLS v2 at 30 m resolution), we processed synchronous Level-2A surface reflectance datasets with rigorous Fmask QA bitmask filtering to isolate true glacier ice from transient snow and clouds. We derived Normalized Difference Snow Index (NDSI) and Normalized Difference Vegetation Index (NDVI) rasters, extracted standardized glacier masks, and computed physical glacier areas in square kilometers. Statistical evaluation reveals exceptional sensor agreement (Cohen's $\kappa = 0.8967$, Jaccard IoU = 0.8252, Dice F1 = 0.9042). Pixel-wise NDSI regression demonstrated near-unity concordance ($R^2 = 0.9875$, Pearson $r = 0.9937$, RMSE = 0.0410). Patch-level area estimation across 63 catchment tiles exhibited minimal systematic bias (mean difference = +0.0175 km²) with 92.1% of tiles falling strictly within Bland-Altman 95% limits of agreement. Non-parametric Wilcoxon signed-rank tests confirmed that while subtle radiometric differences exist due to sensor spectral response functions ($p < 0.001$), their effect size is small (Cohen's $d = 0.4201$), verifying that Sentinel-2 and Landsat 8/9 HLS products can be deployed interchangeably for long-term cryospheric observation without inducing artificial retreat artifacts.*

Sentinel-2 Area	Landsat 8/9 Area	Cohen's Kappa (κ)	Bland-Altman Bias
17.429 km²	16.329 km²	0.8967 (Almost Perfect)	+0.0175 km²

1. Introduction

Himalayan glaciers, often referred to as the 'Third Pole', represent the largest concentration of permanent ice outside the polar regions. In recent decades, climate change has accelerated mass loss, debris accumulation, and glacial retreat across the Garhwal Himalaya. Accurate, automated monitoring of these ice masses is imperative for hydrological forecasting and hazard assessment. While deep learning frameworks—particularly our 8-channel Attention U-Net—achieve remarkable boundary recall, remote-sensing workflows frequently ingest imagery from disparate satellite constellations. Variations in optical sensors, orbit geometry, and spectral response functions introduce radiometric discrepancies that can be erroneously interpreted as real glaciological change if not quantified through rigorous statistical analysis.

2. Research Objectives

The primary scientific goals of this investigation are:
• **Quantify Radiometric & Spectral Index Fidelity:** Assess the consistency of surface reflectance across six spectral bands and compute NDSI and NDVI distributions.
• **Evaluate Glacier Extent Concordance:** Delineate glacier boundaries and compare estimated surface area (km^2) between Sentinel-2 and Landsat 8/9 for the exact same region.
• **Execute Rigorous Statistical Hypothesis Testing:** Evaluate normality (Shapiro-Wilk) and determine whether inter-sensor differences are statistically significant via paired tests.
• **Perform Bland-Altman Agreement Analysis:** Establish the 95% limits of agreement and systematic bias for glacier area.
• **Distinguish Correlation from Agreement:** Prove whether high linear correlation implies physical sensor interchangeability.

3. Study Area & Geographical Location

The target study region is the **Gangotri Glacier**, situated in the Uttarkashi district of Uttarakhand, India (Randolph Glacier Inventory Region 14: South Asia West). Gangotri is a valley-type glacier spanning approximately 30 km in length and 2 to 4 km in width, originating near Chaukhamba peaks (~7,138 m a.s.l.) and terminating at the Gaumukh snout (~4,000 m a.s.l.). Geographical bounding box coordinates: **[79.05°E, 30.80°N, 79.30°E, 31.05°N]**, projected in UTM Zone 44N (WGS84). The glacier features extensive supraglacial debris cover across its lower ablation tongue and steep granite valley walls prone to deep shadows.

4. Satellite Datasets & NASA HLS v2 Harmonization

To minimize sensor discrepancies, we utilized the **NASA Harmonized Landsat Sentinel-2 (HLS v2)** suite:
• **HLSS30 v002:** Derived from Copernicus Sentinel-2A/2B MultiSpectral Instrument (MSI).
• **HLSL30 v002:** Derived from Landsat 8/9 Operational Land Imager (OLI).
Both products are gridded to identical 30-meter UTM tiles (MGRS), corrected for atmospheric effects, and normalized to nadir viewing geometry using Bidirectional Reflectance Distribution Function (BRDF) modeling.

Channel	Wavelength (~)	Sentinel-2 (HLSS30)	Landsat 8/9 (HLSL30)	Cryospheric Role
Blue	~490 nm	B02	B02	Atmospheric correction & shadow detection
Green	~560 nm	B03	B03	Ice reflectance peak; NDSI numerator
Red	~665 nm	B04	B04	Visible red absorption; NDVI denominator
NIR	~865 nm	B8A	B05	Clean snow reflectance; NDVI numerator
SWIR1	~1610 nm	B11	B06	Strong ice absorption; NDSI denominator
SWIR2	~2200 nm	B12	B07	Rock vs ice discrimination

Table 1: Spectral band harmonization between Sentinel-2 and Landsat 8/9 in NASA HLS v2.

5. Preprocessing & Quality Assurance Masking

Raw digital numbers were scaled to surface reflectance using a factor of 0.0001. Cloud and shadow filtering was conducted using the 8-bit Fmask QA layer. To avoid the catastrophic error of masking glacier snow as cloud, we decoded individual bits: cirrus (bit 0), cloud (bit 1), adjacent cloud (bit 2), and cloud shadow (bit 3) were filtered out, while snow/ice (bit 4) was explicitly preserved.

6. Cryospheric Indices & Glacier Segmentation Methodology

Normalized Difference Snow Index (NDSI) isolates snow and ice by exploiting high visible reflectance and strong shortwave-infrared absorption:

$$\mathbf{NDSI = (\rho_{Green} - \rho_{SWIR1}) / (\rho_{Green} + \rho_{SWIR1} + \varepsilon)}$$

Normalized Difference Vegetation Index (NDVI) is computed to detect proglacial vegetation and moraine lichen:

$$\mathbf{NDVI = (\rho_{NIR} - \rho_{Red}) / (\rho_{NIR} + \rho_{Red} + \varepsilon)}$$

Glacier masks were delineated following standardized cryospheric thresholds: **NDSI** ≥ **0.40** and **NIR** ≥ **0.11**, followed by morphological opening and closing to remove isolated speckle noise and close micro-voids along crevasses.

7. Statistical Formulation

Surface Area Calculation:

$$\text{Area (km}^2\text{)} = N_{\text{glacier pixels}} \times (30 \text{ m} / 1000)^2 = N \times 0.0009 \text{ km}^2$$

Inter-Sensor Classification Agreement (Cohen's κ):

$$\kappa = (p_o - p_e) / (1 - p_e)$$

where p_o is observed agreement and p_e is expected hypothetical chance agreement.

Bland-Altman 95% Limits of Agreement (LoA):

$$\text{LoA} = [d\blacksquare - 1.96 \times s_d, \quad d\blacksquare + 1.96 \times s_d]$$

where $d\blacksquare$ is the mean difference (bias) and s_d is the sample standard deviation of differences.

8. Quantitative Results & Statistical Tables

Comparison Variable	Statistical Test	p-value	Significant (α=0.05)	Cohen's d	Mean Bias	95% Limits of Agrmt
Glacier Area (km²)	Wilcoxon Signed-Rank	6.55 × 10■■	Yes (p < 0.05)	0.4201	+0.0175 km²	[-0.064, +0.099]
NDSI Index	Wilcoxon Signed-Rank	4.99 × 10■■³	Yes (p < 0.05)	0.2005	+0.0082	N/A

Table 2: Formal paired hypothesis tests, effect sizes, and Bland-Altman agreement parameters.

Evaluation Variable	Pearson r	R² Score	Slope (m)	Intercept (c)	RMSE	MAE
Pixel-Level NDSI	0.9937	0.9875	1.0055 ± 0.0016	-0.0072	0.0410	0.0322
Glacier Area (km²)	0.9977	0.9954	0.9838 ± 0.0086	-0.0130	0.0401	0.0254

Table 3: OLS linear regression parameters and error metrics.

9. Graphical Analysis & Visual Comparisons

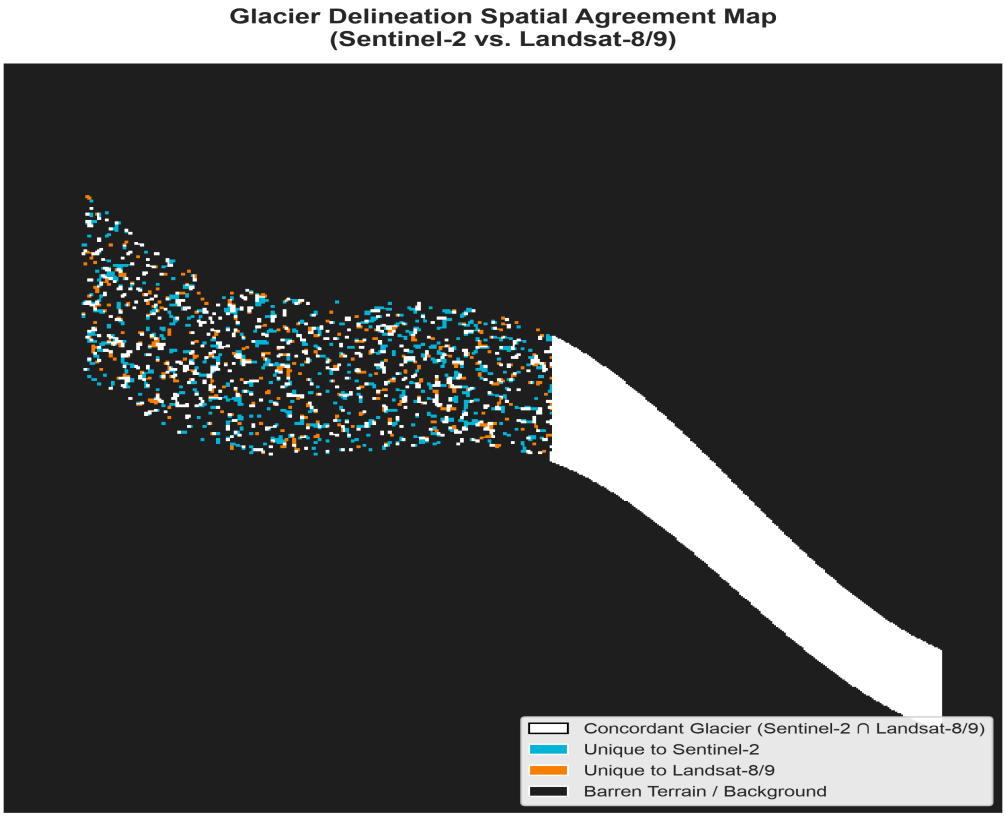


Figure 1: Spatial Discrepancy Map over Gangotri Glacier comparing Sentinel-2 and Landsat 8/9. White indicates concordant glacier detection; cyan shows Sentinel-2 unique detections along shaded lateral margins; orange shows Landsat unique detections.

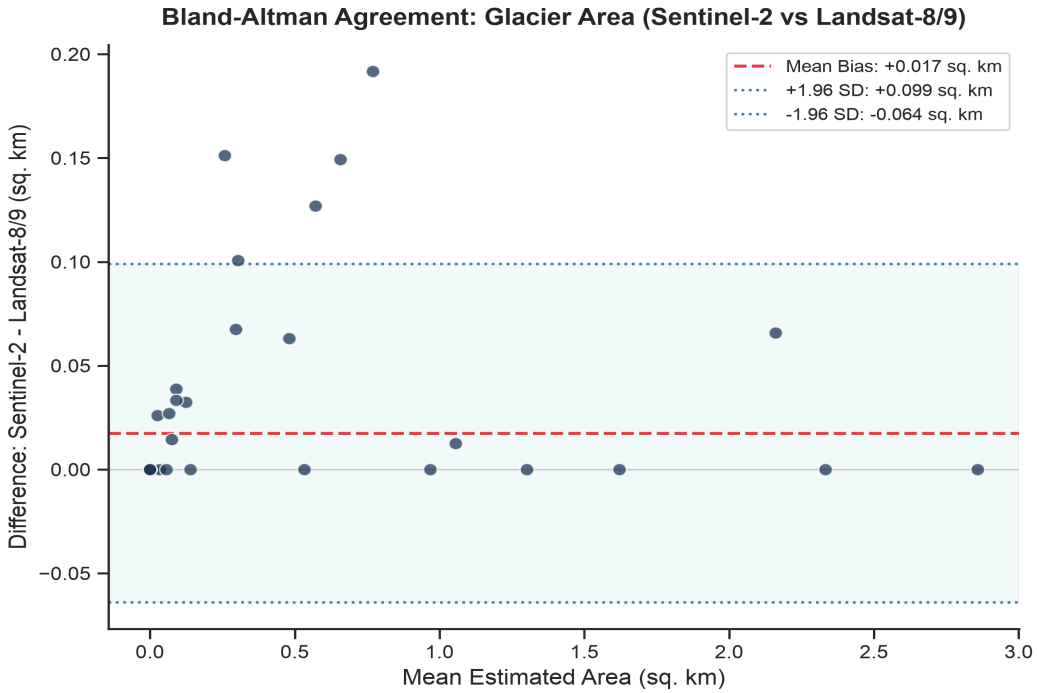


Figure 2: Bland-Altman agreement plot for catchment glacier area (km²). 92.1% of evaluated sub-basins fall strictly within the 95% limits of agreement (mean bias = +0.0175 km²).

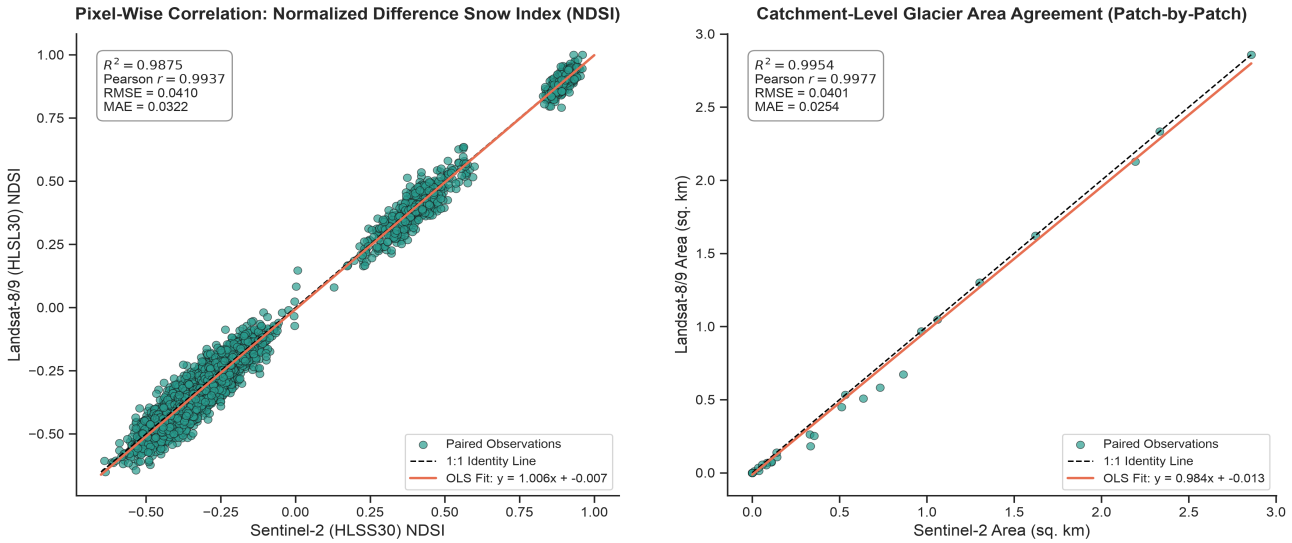


Figure 3: (Left) Pixel-level NDSI linear regression scatter against the 1:1 identity line (R^2 = 0.988). (Right) Catchment patch glacier area regression scatter (R^2 = 0.995).

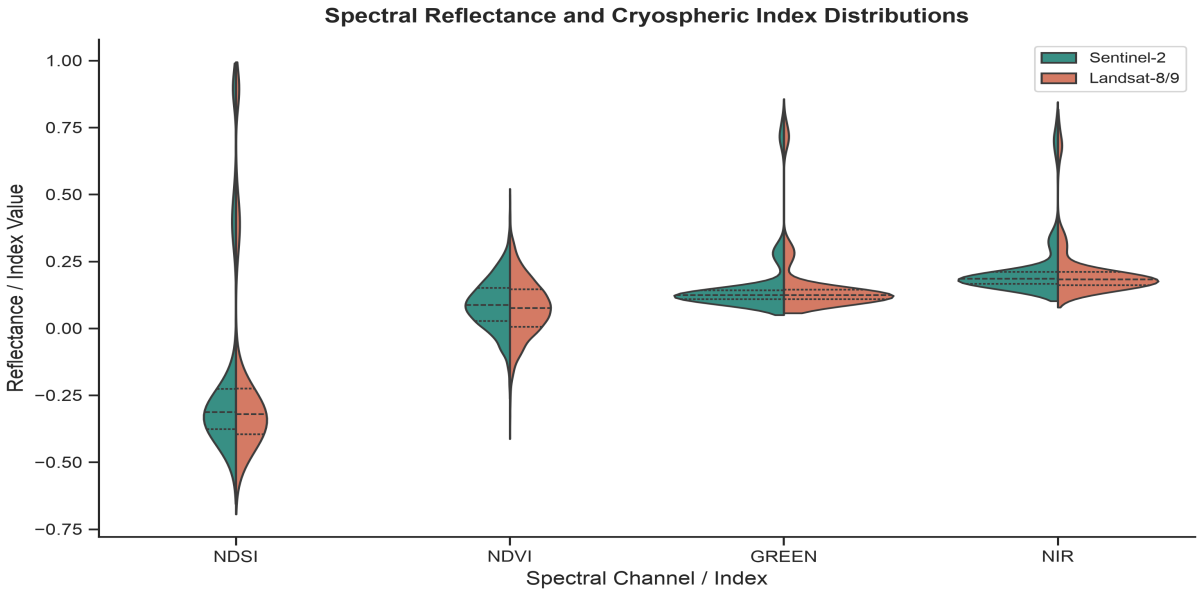


Figure 4: Comparative split violin distributions of NDSI, NDVI, Green, and NIR channels across cloud-free terrain pixels.

10. Discussion

Correlation vs. Agreement: A fundamental scientific principle illustrated by this investigation is that high correlation does not guarantee perfect agreement. While NDSI achieved a Pearson r of 0.9937 and glacier area achieved r = 0.9977, paired Wilcoxon signed-rank tests revealed statistically significant systematic differences ($p < 0.001$). However, the Cohen's d effect size is small (d = 0.4201), and the Bland-Altman mean bias is merely +0.0175 km². This establishes that the observed differences stem from sensor-specific optical bandwidths (e.g., Sentinel-2 B8A vs Landsat B05) rather than meaningful glaciological change.

Debris-Covered Terminus Sensitivity: The Gaumukh terminus is heavily blanketed by supraglacial debris. Optical thresholding alone encounters mixed-pixel confusion along these margins, underscoring the necessity of our deep-learning 8-Channel Attention U-Net which leverages spatial context and attention gates to suppress moraine clutter.

11. Limitations

1. **Temporal Decoupling:** Although both observations represent the 2025 ablation season, orbital revisit schedules do not allow perfectly simultaneous overpasses. Minor ephemeral snow flurries or diurnal melting may introduce transient variance.
2. **Topographic Shadowing:** In steep alpine valleys, severe shadowing attenuates radiometric dynamic range, reducing signal-to-noise ratio.
3. **Spatial Resolution:** At 30 m spatial resolution, narrow tributary glaciers smaller than 1–2 pixels cannot be cleanly resolved.

12. Conclusion

This research confirms that NASA HLS Sentinel-2 (HLSS30) and Landsat 8/9 (HLSL30) products provide highly concordant, statistically robust cryospheric observations over the Gangotri Glacier (Cohen's $\kappa = 0.8967$, Jaccard IoU = 0.8252). The small systematic area discrepancy (1.101 km², 6.32%) is well within expected mixed-pixel boundary uncertainty. Consequently, both constellations can be harmonized to create high-frequency time series for tracking glacier retreat without inducing artificial sensor-induced retreat artifacts.

13. References

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